

The striking irony here is that the last possible object that can be linked to the German currency is precisely such a curve: the reichsmark (as the currency was previously called) went from four per dollar to *four trillion* per dollar in the space of a few years during the 1920s, an outcome that tells you that the bell curve is meaningless as a description of the randomness in currency fluctuations. All you need to reject the bell curve is for such a movement to occur once, and only once—just consider the consequences. Yet there was the bell curve, and next to it Herr Professor Doktor Gauss, unprepossessing, a little stern, certainly not someone I'd want to spend time with lounging on a terrace, drinking pastis, and holding a conversation without a subject.

Shockingly, the bell curve is used as a risk-measurement tool by those regulators and central bankers who wear dark suits and talk in a boring way about currencies.

The Increase in the Decrease

The main point of the Gaussian, as I've said, is that most observations hover around the mediocre, the average; the odds of a deviation decline faster and faster (exponentially) as you move away from the average. If you must have only one single piece of information, this is the one: the dramatic increase in the speed of decline in the odds as you move away from the center, or the average. Look at the list below for an illustration of this. I am taking an example of a Gaussian quantity, such as height, and simplifying it a bit to make it more illustrative. Assume that the average height (men and women) is 1.67 meters, or 5 feet 7 inches. Consider what I call a *unit of deviation* here as 10 centimeters. Let us look at increments above 1.67 meters and consider the odds of someone being that tall.*

10 centimeters taller than the average (i.e., taller than 1.77 m, or 5 feet 10): 1 in 6.3

20 centimeters taller than the average (i.e., taller than 1.87 m, or 6 feet 2): 1 in 44

30 centimeters taller than the average (i.e., taller than 1.97 m, or 6 feet 6): 1 in 740

40 centimeters taller than the average (i.e., taller than 2.07 m, or 6 feet 9): 1 in 32,000

50 centimeters taller than the average (i.e., taller than 2.17 m, or 7 feet 1): 1 in 3,500,000